

Cross-Track Collaboration — Week 5

Track: Data Analytics

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HealthConnect is a shared project, so Week 5 asks each track to identify at least one dependency or collaboration point with another track. This is a simulated handoff: I have worked out what the Data Science track will need from the Data Analytics work and prepared my outputs for it, rather than exchanging files with a specific person.

1. Which track

Data Science.

2. The dependency, and what is being handed over

The Data Science track will build a predictive model on the same 5,000-row appointment dataset. Their work depends on decisions the Data Analytics track makes first, in particular the definition of the outcome variable. The handover from my Week 5 work covers three things:

- **The outcome-variable recommendation.** Keep `appointment_outcome` as three classes (Attended, No-Show, Cancelled) and do not merge Cancelled into No-Show. Section 5 of the notebook (KPI 5) gives the evidence: the cancellation-to-no-show ratio moves in a clear direction as booking lead time grows (0.202 down to 0.076), which means the two outcomes behave differently and should not be collapsed into one target.
- **Three validated candidate features.** `previous_no_shows`, `booking_lead_days`, and `distance_to_clinic_km`. Section 4 of the notebook shows they are independent of each other (lead time and distance correlate near zero, and each effect holds up when checked inside the others' groups), so the Data Science track can treat all three as useful features without first testing for redundancy.
- **The processed dataset and the engineered bands.** `data/processed/healthconnect_appointments_prepared.csv`, with the documented boundaries for `prior_no_show_bracket`, `lead_time_band`, and `distance_band`, so feature engineering can start from there instead of the raw CSV.

3. Why the dependency matters

Week 4 already flagged this as a risk. If the Data Science track picked a different prediction target, for example by merging Cancelled into No-Show, before the outcome categories were checked, the two tracks' work would drift apart and the KPI framing here would no longer line up with their model. The outcome-variable decision needs to be settled with evidence before they lock a target for their baseline model.

4. What changed in my Week 5 work as a result

Working out what the Data Science track would need shaped how I did the analysis:

- I kept the three-class outcome throughout and built KPI 5 specifically so the "keep them separate" recommendation rests on evidence rather than the Week 4 argument alone.
- I ran the confounding check in Section 4 partly so the three features could be handed over as genuinely independent, not just individually interesting.
- I saved the processed dataset and documented every band boundary so the handover is a file and a short spec, not something the Data Science track has to reconstruct from the notebook.

If a Data Science intern is available before the model is built, the next step is a short review of the outcome-variable recommendation with them so it is agreed rather than assumed.